



The Effectiveness of Integrated Management Packages in Controlling Aphid-Borne Virus Infections on Groundnut in South Sulawesi, Indonesia

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ABSTRACT

Aphid-borne viral infection is a major limiting factor for groundnut production in South Sulawesi, Indonesia. The study aimed to elucidate the effectiveness of integrated management packages (IMP) in controlling aphid-borne viruses infecting groundnut in the field. Field trials were conducted in a plot without (M0) and with plastic mulch and maize as a plant barrier (M1), while two different groundnut varieties, *Bougainvillea* leaf extract, and insecticides were incorporated as the subplot treatments. The treatments in M1 plot significantly reduced the aphid population (17.4–79.9%) and AUDPC incidence (51.7–87.3%), and severity (61.0–92.9%), respectively. The treatments increased dry pod production by 11.9–63.3% depending on subplot treatment compared to treatments in the M0 plot ($p > 0.05$). *Bean common mosaic virus* strain PST and *Cowpea Polerovirus* 2 were positively detected in groundnuts as well as in aphids using RT-PCR. Further, nine weed species in the vicinity of the field trial were found to be positively infected with BCMV-PST, indicating their crucial role as reservoir hosts. Interestingly, the effectiveness of *Bougainvillea* leaf extract treatment in the M1 plot was comparable to that of the insecticide treatment, indicating that this IMP is a more cost-effective, eco-friendly groundnut virus control than the currently used method.

INTRODUCTION

Groundnut (*Arachis hypogaea* L.) is a member of the fabaceae and ranks as the second most important commodity after soybeans. It serves a strategic role in supplying a source of protein and vegetable oil for national food security. Groundnuts are grown extensively in Indonesia and harvested yearly in monocultures and intercrops as food and industrial crops. South Sulawesi is the second-largest producer of groundnuts outside Java, after West Nusa Tenggara. However, since 2015, groundnut area and production have declined in Indonesia, including in South Sulawesi. In 2020, groundnut production in South Sulawesi had declined by 58% compared to 2016 values (Indonesian Ministry of Agriculture, 2021). The decline in national groundnut productivity is attributed to various factors, including cultivation

techniques, low seed quality, the use of local varieties with low yields, and the occurrence of pest and plant diseases (Sumarno et al., 2000; Kasim & Kadir, 2014; Paturohman & Sumarno, 2014). Pathogen infections such as bacteria, fungi, and viruses in groundnut plants are among the factors that have contributed to the decline in production (Elwakil & El-Metwally, 2001; Saleh & Baliadi, 2015)

Potyvirus, *Cucumovirus*, *Tospovirus*, and *Polerovirus* are genera of viruses reported to infect groundnuts in Indonesia. The *Potyvirus* genus is the largest known cluster of plant RNA viruses, encompassing over 180 species, making up around 15% of all recognized plant viruses (Valli et al., 2021; Wylie et al., 2017). Identified as the founding genus in the Potyviridae family (Dolja et al., 2020), Potyviruses feature a single-stranded (ss) linear

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positive-sense RNA genome of approximately 10 kb. This genome includes a molecule of one of the viral proteins (VPg) covalently attached to its 5' end, along with a 3-terminal poly A tail (Rodamilans et al., 2015).

The BCMV strain Peanut stripe (PSt) and BCMV strain Blackeye (B1C) are members of the *Potyvirus* genus that can be transmitted through seeds (seed-borne transmission) with an efficiency of 80%, depending on the type and variety of plants and aphids (Aphids transmission) (Drijfhout & Morales, 2005), and through mechanical inoculation. There are 36 species of aphids from 21 genera that can serve as vectors and are effective in transmitting BCMV, including *Acyrtosiphon pisum*, *Aphis fabae*, *Myzus persicae*, and *Aphis craccivora* (Wamonje et al., 2020). *Cucumber mosaic virus* (CMV) (Cucumovirus: Bromoviridae) is spread from plant to plant by various species of aphids, such as *Aphis gossypii* and *Myzus persicae*, in a non-persistent manner. Additionally, CMV can be transmitted through seeds, plant sap, mechanical inoculation, and weeds (Lenardon et al., 2002). CMV has been reported to infect peanuts naturally in Argentina, China, and South Korea (de Breuil et al., 2005; Xu & Barnett, 1984; Park et al., 2015). Tospoviruses (family Bunyaviridae, genus *Tospovirus*) rank among the most significant plant virus groups globally, causing infections in a broad spectrum of economically important crops (Mandal et al., 2012). Tospoviruses are quasisphepherical, with a diameter ranging from 80 to 110 nm. They are enveloped isometric RNA viruses with a tripartite genome consisting of small (S), medium (M), and large (L) segments of single-stranded RNA (Bhat et al., 2002). TSWV can be transmitted by specific insect vectors, namely *Thrips tabaci*, *T. setosus*, *Frankliniella* spp., and *Scirtothrips* spp., through circulative propagative means and mechanical transmission. However, it cannot be transmitted through seeds or pollen (Mandal et al., 2001; Peters, 2003). *Cowpea polerovirus 2* (CPPV2) and *Chickpea chlorotic stunt Polerovirus* (CpCSV) are new members of the *Polerovirus* genus. CPPV2 was first reported to infect cowpea plants (*Vigna unguiculata* L.) in Burkina Faso (Palanga et al., 2016), while CpCSV was initially identified as infecting chickpea plants (*Cicer arietinum*) in Ethiopia and has been reported in several countries, including Sudan, Tunisia, and Syria. CpCSV is transmitted by the aphid species *Aphis craccivora* and *Acyrtosiphon pisum* persistently (Abraham et al., 2006), whereas no reported aphid species are acting as vectors for CPPV2.

Bean common mosaic virus strain peanut stripe (BCMV-PSt) has been reported to infect groundnut in South Sulawesi (Saleh & Baliadi, 2015). Further, a recent survey of several central groundnut producers in South Sulawesi province showed that BCMV and *cucumber mosaic virus* (CMV) have been found to infect groundnuts, with prevalence ranging from 30% to 100% based on serological detection by dot-blot immunobinding assay (DIBA). The existence of BCMV strain black eye cowpea (BCMV-BIC), CMV and *Polerovirus* has been confirmed by RT-PCR (Kasim et al., 2022). In addition, BCMV-BIC (Damayanti et al., 2009), *Groundnut bud necrosis virus* (PBNV) (Damayanti & Naidu, 2009), *Soybean mosaic virus* (SMV) (Andyanie, 2012), *Cucumber mosaic virus* strain soybean (CMV-S) (Damayanti & Wiyono 2015), *Cowpea mild mottle virus* (CPMMV) (Kulsum, et al., 2016) are also important viruses infecting legumes in Indonesia. Aphid-borne viruses that infect legumes worldwide include *Peanut mottle virus* (PeMoV) (de Breuil et al., 2015), BCMV-PSt (Mabele & Wotia, 2021), CMV (de Breuil et al., 2012), *Peanut stunt virus* (PSV) (Park et al., 2015) and *Polerovirus* (Palanga et al., 2016). Viral infections can cause significant yield losses and reduce seed quality.

Management of aphid-borne virus(es) commonly involves controlling the vector insects by using insecticides, and by biological control using predators (Saleh, 2007). The effectiveness of chemical insecticides in controlling aphid vectors is unsatisfactory (Hooks & Fereres, 2006). This is because viruses transmitted by aphids in a non-persistent manner will be rapidly transmitted to healthy plants soon after the application of insecticide (before the aphids are killed). Moreover, aphids can develop resistance to commonly used pesticides (Fereres, 2000; Li & Han, 2004). However, the use of barrier crops has been shown to reduce aphid populations in major crop areas (Olubayo et al., 2009). Barrier crops such as maize, sorghum, and sunflower have been reported to effectively decrease the disease incidence of the aphid-borne viruses CMV, *potato virus Y*, and *pepper vein mottle virus* in cayenne pepper and watermelon by 24–46% (Waweru, et al., 2021). Similarly, sorghum barrier plants and pumpkins have demonstrated a 43–96% decrease in the incidence of *watermelon mosaic virus* and *papaya ringspot virus* (Damicone et al., 2007). Previously, *Bougainvillea* sp. leaf extracts were reported to effectively reduce mosaic disease intensity and BCMV titer on yardlong beans

in greenhouse trials (Damayanti & Panjaitan 2014) and utilizing maize as a barrier plant reduced BCMV infection on yardlong beans in the field (Damayanti & Pebriyeni 2015).

In South Sulawesi and other groundnut cultivation areas in Indonesia, management of diseases on groundnuts, including viruses, relies on pesticides. The active ingredients of pesticides commonly used by farmers such as chlorpyrifos, cypermethrin, permethrin, paraquat dichloride, methomyl, carbaryl, BPMC, and mancozeb. Generally, farmers are unaware of and do not recognize symptoms related to viral disease on groundnut. Utilizing local groundnut varieties for generations contributes to aggravating the decline in groundnut production. Furthermore, farmers in this area have a unique habit of reusing plastic mulch from previous vegetable crops to cultivate groundnut in the fields. To increase the effectiveness of reused plastic mulch in controlling aphid-borne viruses infecting groundnut, we combined it with the use of a barrier crop in the plots and compared the treatments of *Bougainvillea* leaf extract and insecticide on subplots of two groundnut varieties.

MATERIALS AND METHODS

Field Trial Preparation

The experiment was conducted from August to November 2020, in a 2,700 m² plot in Maros regency, South Sulawesi province, Indonesia. The main plot treatment consisted of plots without plant barriers or plastic mulch (M0) and plots with maize barriers and plastic mulch (M1), while the subplot treatments consisted of P1: local seed variety + chemical insecticide; P2: local seed variety + *Bougainvillea* leaf extract; P3: Takar 2 seed variety + chemical insecticide; P4: Takar 2 seed variety + *Bougainvillea* leaf extract; and P5: local seed variety (control). Each treatment was replicated in 3 blocks. A local maize variety as a barrier was planted two weeks before groundnut planting and consisted of 2 rows surrounding the main plot treatment (Fig. 1). The active ingredients of the insecticide were Profenofos 300 g/l and lambda-cyhalothrin 50 g/l, which is a combination commonly used by the local farmers.

Preparation of *Bougainvillea* Leaf Extract

Crude leaf extract of *Bougainvillea* sp. was prepared according to the procedure developed by Damayanti & Panjaitan (2014), using a 1:10 (w/v) ratio of leaf and water. Before planting, the groundnut seeds were soaked in the crude *Bougainvillea* leaf extract for 12 hours. Subsequently, the *Bougainvillea*

leaf extract and the insecticide treatment were applied five times at two-week intervals 2–10 weeks after planting (WAP) in either of the main plots.

Observation Variables

The observed variables consisted of the aphid population dynamic, disease assessment parameters such as disease incidence and severity, area under the disease progress curve (AUDPC), detection of viruses infecting groundnuts, weeds, and aphids using RT-PCR, and yield at the end of the experiment. The aphid population dynamic, disease assessment parameters, and yield were observed on thirty sample plants diagonally per treatment plot. The number of pods (seeds), wet pod weight (grams), and dry pod weight (grams) were observed for each sample plant to estimate groundnut production. The fresh plants were dried at 110°C for 4 hours.

Virus Detection Using RT-PCR

Virus detection was carried out on samples of groundnut plants, weeds, and aphids from the experimental field and the surrounding area. Virus detection was conducted on leaf samples obtained from composite treatments in the field (10 composite samples) 12 weeks after planting, weed plants surrounding the groundnut cultivation area (16 samples), and aphids from groundnut samples in the experimental plots (4 composite samples). Total RNA extraction was performed using a CTAB method with minor modification on lysis duration at a temperature of 65°C from 60 min to 30 min (Doyle & Doyle, 1987). RT-PCR was preceded by reverse transcription (RT) to synthesize complementary DNA (cDNA) using the Tetro cDNA Synthesis Kit according to procedures provided by Bioline (USA). The cDNA was amplified by PCR using a universal primer for *Cucumovirus*, and specific primers (Table 1). The PCR reactant was prepared according to the MyHS Taq polymerase protocol (Meridian Bioscience, USA) with a program customized to the target virus. The DNA of the PCR product was separated, visualized, and documented digitally.

Data Analysis

The study was designed as a randomized split-plot design with five management package treatments, and each treatment consisted of three subplots as replications. Disease severity data were tabulated and analysed using non-parametric ANOVA (Friedman test-Realstat). If the treatment had a significant effect, multiple comparison tests were performed using the Dunn test at a 95% confidence level.

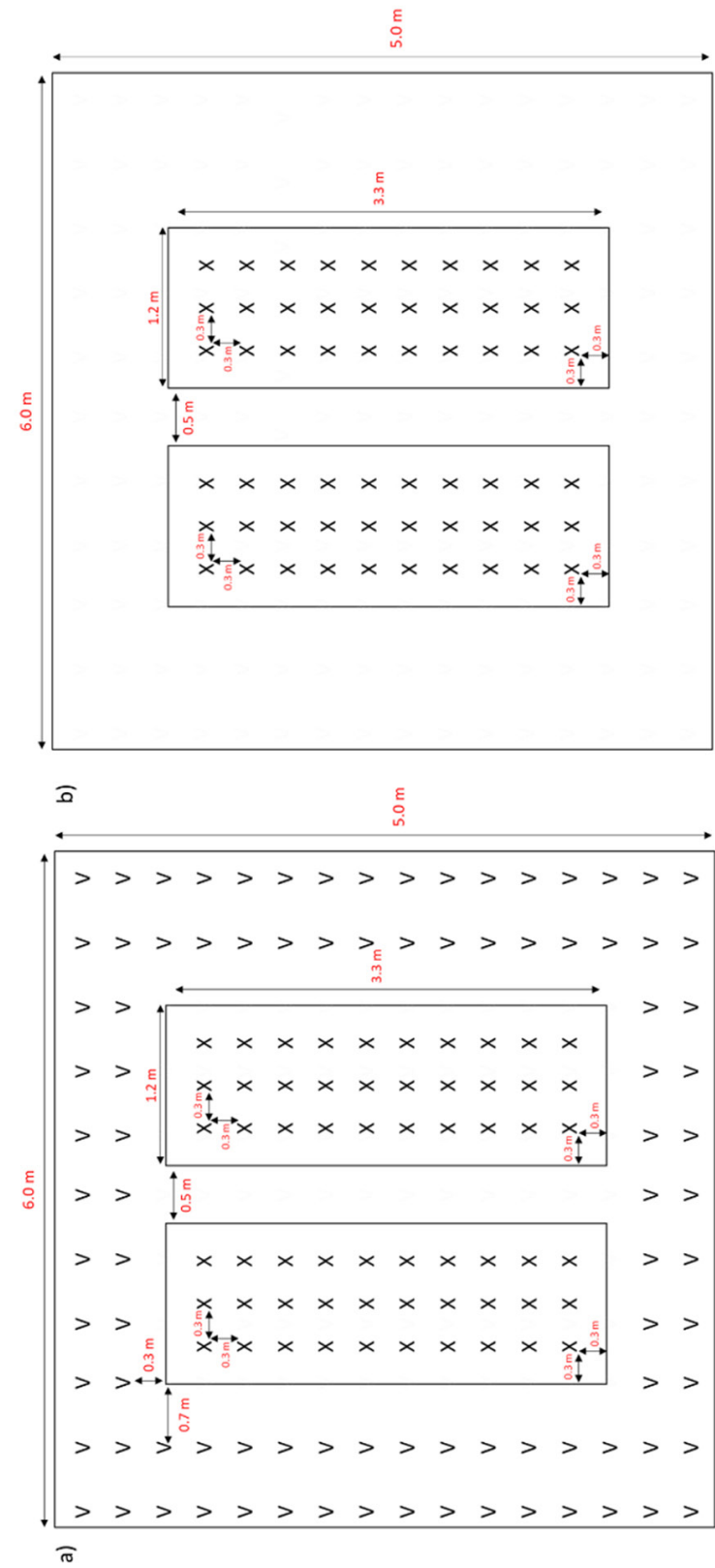


Fig. 1. The layout of the main plot and treatment subplots (a) using maize barrier plants and plastic mulch (M1), (b) without maize barrier plants and plastic mulch (M0). Remarks: x = groundnut plants; v = maize plants

The relationship between aphid population dynamics and disease incidence and severity was analyzed by a simple linear regression. Correlation categories were grouped based on coefficient of determination with R² values as follows: weak (0–40%), moderate (41–60%), and strong (61–100%).

Determination of the effectiveness of each treatment subplot was carried out by comparing each observation variable in the M1 plot with the M0 plot in each treatment with the formula:

$$\text{Effectiveness of treatment} = \frac{\text{Variable treatment in M1} - \text{Variable treatment in M0}}{\text{Variable treatment in M0}} \times 100\%$$

RESULTS AND DISCUSSION

Effect of Treatment on Aphid Population, Disease Incidence, and Disease Severity

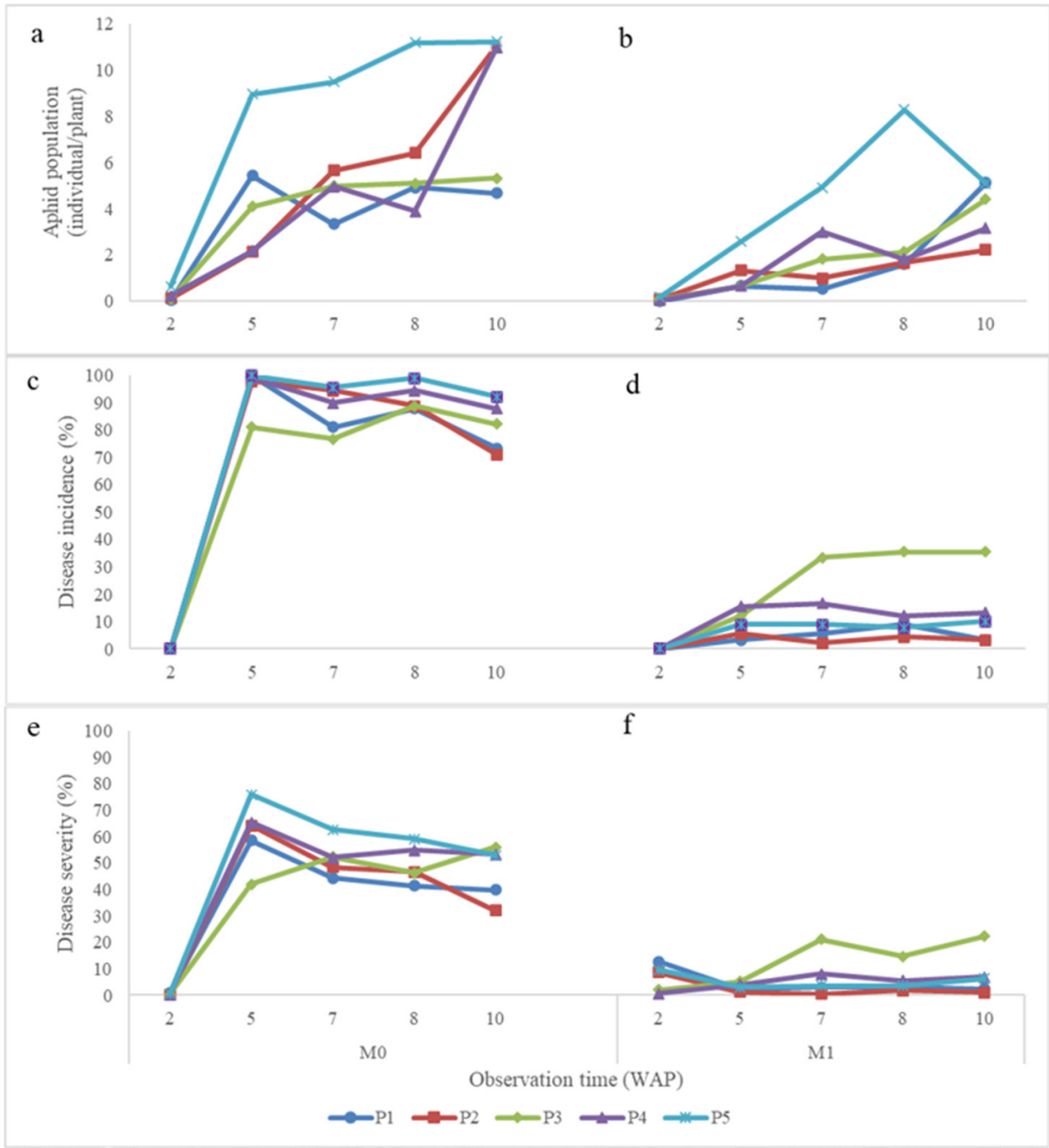
The subplot treatments in plot M1 showed that the aphid population dynamic, the disease incidence, and the disease severity were significantly lower than under similar treatments in plot M0 (Fig. 2a–f; Table 2). Among treatments, M1P2 and M1P4 showed the lowest aphid population compared to the other treatment combinations (Table 2a). The aphid population in M0 was higher than in M1 based on the R² value between treatments and aphid populations, which ranged from 62–97% (Table 3). This indicated that the subplot treatments in M0 and M1 were strongly correlated with the aphid populations. The integrated management packages for controlling the aphid-borne viruses infecting groundnut showed that the incorporated subplot treatments in the M1 plot effectively reduced the population of aphids, with effectiveness ranging from 17.4–79.9% compared to the treatments in the M0 plot, except for M1P1. The highest effectiveness was found in the M1P2 and M1P4

treatments, which showed significant reductions in weekly aphid populations of up to 79.9% and 71.1%, respectively (Table 4). The use of barrier crops (M1) as a component of IMP plays a role in protecting primary crops (in this study, groundnut plants) from insect vectors (aphids). Aphids locate their host plants visually by distinguishing the green colour of the plants' leaves from the soil colour. The use of maize barriers can disguise and divert aphid attention from the primary crop (groundnut) (Batzer et al., 2022). Aphids settling on the barrier plant will try to confirm whether the plant is a suitable host by probing and sucking cellular fluids from the barrier plant during this orientation phase. While probing, the viruliferous aphids lose the virus and become non-viruliferous ones. As a result, the aphids that settle on the main plant will be virus-free; barrier plants act as a sink (Ferreles, 2000).

Disease incidence and severity caused by viral infection were higher in the M0 treatment compared to the M1 plot. The AUDPC values of disease incidence and severity in each combination treatment in the M1 plot were significantly lower than in the M0 plot. The M1P2 combination treatment showed the lowest incidence and severity of disease based on the AUDPC of disease intensity values (Table 2b–c). Generally, the treatments incorporated in the M1 plot were shown to have effectively suppressed the AUDPC of disease incidence by 51.7–87.3%, and disease severity by 61.0–92.9%, compared to similar treatments in the M0 plot (Table 4). This finding proves that managing aphid-borne viruses in groundnut plants by using maize as a barrier, reusing mulching, and applying bougainvillea leaf extract effectively reduced the incidence and severity of viral diseases in the field.

Table 1. List of primer sequences used in RT-PCR detection

Viruses	Primers	Sequence	DNA size (bp)	Reference
<i>Cucumovirus</i>	CPTALL-5'	5'-YASYTTTDRGGTTCAATTCC-3'	~900	Choi et al. (1999)
	CPTALL-3'	5'-GACTGACCATTTTAGCCG-3'		
BCMV-PSTV	PST 1	5'-GCATGCCCTCGCCATTGCAA-3'	234	Higgins et al. (1999)
	PST 2	5'-GCACACACTTCTTGGCATGG-3'		
BCMV-BIC	BIC-cpF	5'-TCAGGAAGTGGGCAGCCGCAAC-3'	861	Melinda et al. (2015)
	BIC-cpR	5'-CTGCGGGGAACCCATGCCAAG-3'		
CPPV2	NB3897F	5'-GAGTTCATCTCCGAGGCC-3'	263	Palanga et al. (2016)
	NB4160R	5'-CDTCTACCTATTTSGGRTTHTG-3'		
CpCSV	CpCSV-F	5-TAGGCGTACTGTTTCAGCGGG-3	413	Najar et al. (2011)
	CpCSV-R	5-TCCTTTGTCCATTCGAGGTGA-3		



Remarks: M0: main plot without mulch and barrier crops, and M1: main plot using mulch and barrier crops. Subplot treatments: P1: local seed variety + chemical insecticides; P2: local seed variety + Bougainvillea leaf extract; P3: Takar 2 seed variety + chemical insecticides; P4: Takar 2 seed variety + Bougainvillea leaf extract; and P5: control (local seed variety). WAP: Weeks after planting

Fig. 2. Effect of treatment on: (a–b) aphid population, (c–d) disease incidence, and (e–f) disease severity

Table 2. Effect of treatment on the aphid population, and viral disease incidence and severity in the field

Treatment	Effect of treatment on the aphid population, virus disease incidence and severity (week after treatment)																
	a. Aphid population							b. Disease Incidence (%)							c. Disease Severity (%)		
	2	5	7	8	10	2	5	7	8	10	AUDPC	2	5	7	8	10	AUDPC
M0P1	0.03 ^{bc}	5.4 ^b	3.3 ^{bc}	4.9 ^c	4.7 ^b	4.4 ^b	100 ^a	81.1 ^b	87.8 ^a	73.3 ^b	161.3 ^a	0.8 ^{bcd}	58.6 ^{bc}	44.2 ^{bc}	41.3 ^b	39.8 ^{bcd}	286.0 ^{ab}
M0P2	0.1 ^{bc}	2.1 ^{cd}	5.7 ^b	6.4 ^b	11.1 ^a	0.0 ^b	97.8 ^a	94.5 ^a	88.9 ^a	71.1 ^b	179.0 ^a	0.0 ^d	64.2 ^{ab}	48.4 ^b	46.7 ^b	32.0 ^{cd}	320.6 ^a
M0P3	0.1 ^{bc}	4.1 ^{bc}	5.0 ^b	5.1 ^b	5.3 ^b	1.1 ^b	81.1 ^a	76.7 ^b	88.9 ^a	82.2 ^{ab}	164.7 ^a	0.2 ^{cd}	42.0 ^{cd}	52.2 ^b	46.2 ^{bc}	56.0 ^{bc}	327.3 ^a
M0P4	0.2 ^b	2.2 ^{cd}	5.0 ^b	3.9 ^{dc}	11.0 ^a	1.1 ^b	98.9 ^a	90.0 ^{ab}	94.4 ^a	87.8 ^a	197.0 ^a	0.2 ^{cd}	65.3 ^{bc}	52.0 ^b	54.9 ^{ab}	53.3 ^{ab}	389.1 ^a
M0P5	0.6 ^a	9.0 ^a	9.5 ^a	11.2 ^a	11.2 ^a	2.2 ^b	100 ^a	95.5 ^a	98.9 ^a	92.2 ^a	202.5 ^a	0.4 ^{bcd}	75.8 ^a	62.7 ^a	59.1 ^a	52.9 ^a	424.8 ^a
M1P1	0.0 ^c	0.7 ^d	0.5 ^d	1.6 ^d	5.1 ^b	47.8 ^a	3.3 ^c	5.5 ^d	8.9 ^{bc}	3.3 ^d	50.0 ^b	12.8 ^a	2.4 ^{ef}	2.9 ^{ef}	2.9 ^d	2.4 ^e	58.7 ^c
M1P2	0.1 ^{bc}	1.3 ^d	1.0 ^{cd}	1.7 ^d	2.2 ^b	36.7 ^a	5.6 ^{bc}	2.2 ^d	4.4 ^c	3.3 ^d	22.7 ^b	8.6 ^a	1.3 ^f	0.4 ^h	1.8 ^d	1.1 ^e	22.7 ^c
M1P3	0.1 ^{bc}	0.7 ^d	1.8 ^{cd}	2.1 ^d	4.4 ^b	6.7 ^b	12.2 ^{bc}	33.4 ^c	35.6 ^b	35.6 ^c	79.5 ^b	2.0 ^b	5.1 ^{ef}	21.1 ^{cd}	14.7 ^c	22.2 ^d	127.7 ^{bc}
M1P4	0.03 ^{bc}	0.7 ^d	3.0 ^{abc}	1.8 ^d	3.2 ^b	0.0 ^d	15.6 ^b	16.7 ^d	12.2 ^{bc}	13.3 ^d	40.7 ^b	0.6 ^{bc}	4.0 ^{de}	8.0 ^{de}	5.6 ^d	6.9 ^e	63.2 ^c
M1P5	0.2 ^{bc}	2.6 ^{cd}	4.9 ^b	8.3 ^b	5.1 ^b	43.3 ^a	8.9 ^{bc}	8.9 ^d	7.8 ^c	10.0 ^d	54.8 ^b	9.7 ^a	2.9 ^{ef}	3.6 ^{ef}	3.6 ^d	6.2 ^e	70.4 ^c

Remark: M0. main plot without mulch and barrier crops, and M1. main plot using mulch and barrier crops. Subplot treatments: P1. local seed variety + chemical insecticides; P2. local seed variety + Bougainvillea leaf extract; P3. Takar 2 seed variety + chemical insecticides; P4. Takar 2 seed variety + Bougainvillea leaf extract; and P5. control (local seed variety). WAP. Weeks after planting; AUDPC. Area under the disease progress curve

Incorporating bougainvillea leaf extract treatment in the M1 plot (M1P2 and M1P4) showed the highest reductions in aphid population, disease severity, and disease incidence compared to other treatments in the M0 plot (Table 4). Our results suggest that the antiviral substances in Bougainvillea leaf extract can potentially be used to manage plant viruses, as they act as biopesticides. *Bougainvillea Antiviral Protein* (BAP) in the plant is a form of induced systemic resistance (ISR). ISR can be induced by weak pathogens (avirulent), botanicals (plant extracts), and environmental stress. ISR induces plants to produce a pathogen inhibitor (Zeller, 2006; Verma et al., 1998). Bougainvillea leaf extract has been reported to reduce virus titre and severity of *squash mosaic virus* (Putri & Damayanti, 2020), reduce the severity of *zucchini yellow mosaic virus* (ZYMV) infection by mechanical inoculation (Güller et al., 2018), prolong incubation, and reduce BCMV and *Peanut mottle virus* (PMoV) disease incidence and severity as well as virus titre in greenhouse (Kurnianingsih & Damayanti, 2012; Damayanti & Panjaitan, 2014; Zulfi et al., 2024). Incorporation of barrier plants and bougainvillea leaf extract is also able to reduce disease intensities and virus titre of BCMV on yard-long beans (Putra, 2014) as well as disease intensity aphid-borne viruses and aphid population on soybean in the fields (Cahyati et al., 2023). Besides that, *Bougainvillea x buttiana* plant extract has been found to produce defense-related proteins (DRPs) that effectively suppress and inhibit the replication of *tobacco mosaic virus* (TMV) (Chaudhary et al., 2008). Various plant

extracts containing secondary metabolites such as alkaloids, anthocyanins, carotenoids, flavonoids, phenolic compounds, essential oils, polysaccharides, and proteins have been shown to possess antiviral characteristics, which are significant in combating both biotic and abiotic stresses (Wink, 2015).

Managing aphid-borne viruses through barrier plants, natural enemies, resistant cultivars, and insecticides is most effective method to control aphids as mosaic disease vectors (Barzman et al., 2015). Combining cultivation techniques with biological and chemical management has proven effective in controlling and reducing losses due to insect infestation (Torres & Bueno, 2018; Biondi et al., 2018). Zhang et al., (2020) also reported that combining management techniques, such as barrier plants and intercropping, is the best alternative for controlling *Bemisia tabaci* in China.

Groundnut Production

Disease incidence and severity generally affected groundnut productivity. The groundnut field production showed an increase under the combination treatment (M1) compared to M0. Groundnut plants under the M1 treatment showed an increase in the number of pods produced, but this was not significantly different from the number of pods produced in M0. The highest number of pods was observed in the M1P4 treatment. Furthermore, significant differences were observed in pod fresh weight and pod dry weight between M1 and M0. The M1P1 treatment resulted in the highest wet and dry pod weights produced (Table 5).

Table 3. Correlation of treatments with the aphid population dynamics

Treatments	M0		M1	
	Regression equation	R²	Regression equation	R²
P1	y = 20.50x - 34	0.77	y = 16.85x - 55.1	0.73
P2	y = 44.40x - 113.6	0.97	y = 9x - 27.4	0.89
P3	y = 22.65x - 41.9	0.83	y = 15.40x - 23	0.62
P4	y = 36.60x - 112.8	0.78	y = 11.15x - 20.3	0.91
P5	y = 39.60x - 32.2	0.87	y = 26.15x - 37.1	0.64

Table 4. Effectiveness of treatment with a combination of maize barriers and plastic mulch (M1) to control aphid-borne viruses in groundnut

Treatment	Aphid population (%)	AUDPC Incidence (%)	AUDPC Severity (%)	Increasing Groundnut production (%)		
				Number of Pod	Wed Pod	Dry pod
M1P1	9.2	-69.0	-79.5	22.67	45.71	63.35
M1P2	-79.9	-87.3	-92.9	6.12	9.71	11.94
M1P3	-17.4	-51.7	-61.0	21.80	31.44	44.01
M1P4	-71.1	-79.3	-83.8	17.54	14.59	19.08
M1P5	-54.6	-72.9	-83.4	11.81	16.78	13.96

Remark: Minus (-) indicates the treatment caused a decrease in the variables compared to the treatment in the M0 plot

Generally, treatment M1 was shown to effectively increase the number of pods from 6.1–22.7%, the weight of wet pods from 9.7–45.7%, and the weight of dry pods from 11.9–63.4% compared to M0 (Table 4). This shows that the treatment involving the plastic mulch and maize barrier (M1) in all sub-treatments effectively increased groundnut production for both groundnut varieties.

The use of plastic mulch is an effective alternative for increasing groundnut production, as it improves water use efficiency, prevents soil erosion, and reduces weed competition. A study by Sun et al. (2015) reported that using black-clear-black-coloured mulch in groundnut cultivation resulted in a 12–14% increase in groundnut production compared to cultivation with transparent plastic mulch. Black-silver plastic mulch has properties similar to glass, as it can reflect sunlight. This reflection can increase photosynthate production, thus positively affecting plant growth according to Nurbaiti et al. (2017).

Black plastic mulch can also act as a heat sink and maintain soil temperature stability. Kusumasiwi et al. (2012) also suggested that plastic mulch can reduce evaporation and meet plants' water needs, thus induce for better root development and growth.

Detection of Virus-like Symptoms on Weeds in the Vicinity of the Field Trial by RT-PCR

Several weed species, including *Ageratum conyzoides* L. (Asteraceae), *Arachis pintoi* Krapov. & W.C. Greg. (Fabaceae), *Ludwigia octovalvis* (Jacq.) P.H. Raven (Onagraceae), *Acalypha indica* L. (Euphorbiaceae), *Melochia corchorifolia* L. (Malvaceae), *Synedrella nodiflora* L. (Asteraceae), *Erechtites valerianifolia* (Link ex Wolf) Less. ex DC. (Asteraceae), *Occimum gratissimum* L. (Lamiaceae), and *Asystasia gangetica* ssp. *micrantha* (Nees) Ensermu (Acanthaceae), showed virus-like symptoms such as yellowing, mosaic, vein clearing, and vein banding, as well as no symptoms (Table 6; Fig. 3).

Table 5. Effect of integrated management packages (IMP) on groundnut yields

Treatment	Average		
	Number of Pods	Wet Pods	Dry Pods (g)
M0P1	24.7±2.8 ^a	56.0±4.2 ^{bc}	41.2±7.2 ^b
M0P2	29.4±5.7 ^a	64.9±7.0 ^{abc}	49.4±8.8 ^b
M0P3	26.6±1.5 ^a	49.3±8.7 ^c	38.4±2.8 ^b
M0P4	28.5±5.4 ^a	56.2±13.0 ^{bc}	43.5±14.3 ^b
M0P5	27.1±2.9 ^a	59.0±11.4 ^{bc}	48.7±5.1 ^b
M1P1	30.3±5.6 ^a	81.6±12.9 ^a	67.3±10.2 ^a
M1P2	31.2±3.1 ^a	71.2±15.1 ^{ab}	55.3±12.2 ^{ab}
M1P3	32.4±7.7 ^a	64.8±15.4 ^{abc}	55.3±15.3 ^{ab}
M1P4	33.5±4.4 ^a	64.4±10.2 ^{abc}	51.8±3.6 ^{ab}
M1P5	30.3±2.9 ^a	68.9±3.5 ^{abc}	55.5±2.1 ^{ab}

Remarks: M0: main plot without mulch and barrier crops, and M1: main plot using mulch and barrier crops. Subplot treatments: P1: local seed variety + chemical insecticides; P2: local seed variety + Bougainvillea leaf extract; P3: Takar 2 seed variety + chemical insecticides; P4: Takar 2 seed variety + Bougainvillea leaf extract; and P5: control (local seed variety)

Table 6. Virus-like symptoms of weeds in the vicinity of the groundnut experiment.

Weed species	BCMV Symptom				
	Yellowing	Mosaic	Vein clearing	Vein banding	No symptom
<i>Ageratum conyzoides</i>	✓	✓		✓	
<i>Arachis pintoi</i>		✓		✓	
<i>Ludwigia octovalvis</i>	✓	✓	✓		
<i>Acalypha indica</i>					✓
<i>Melochia corchorifolia</i>					✓
<i>Synedrella nodiflora</i>					✓
<i>Erechtites valerianifolia</i>					✓
<i>Occimum gratissimum</i>				✓	
<i>Asystasia gangetica</i> s. sp. <i>micrantha</i>		✓	✓	✓	

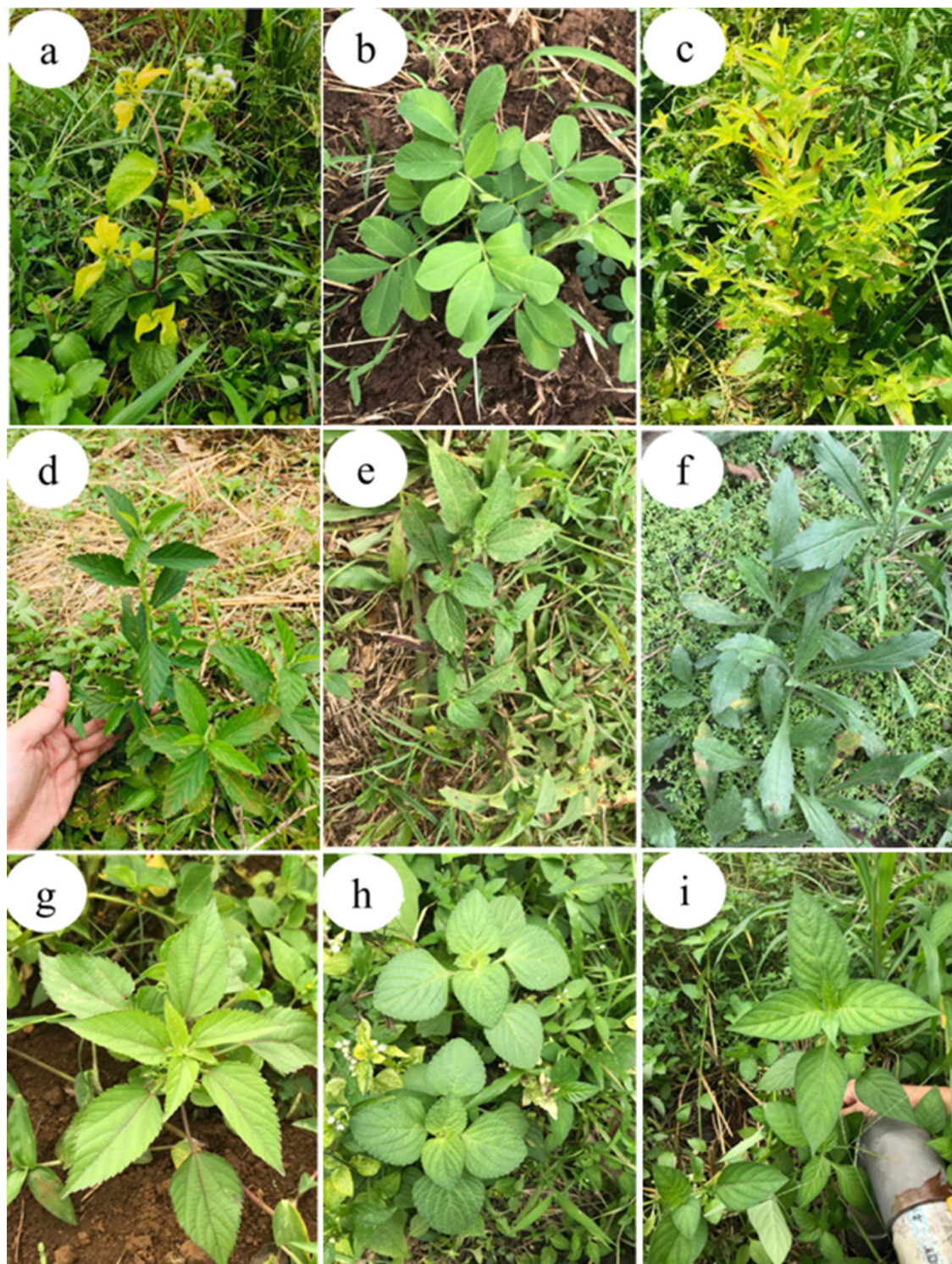
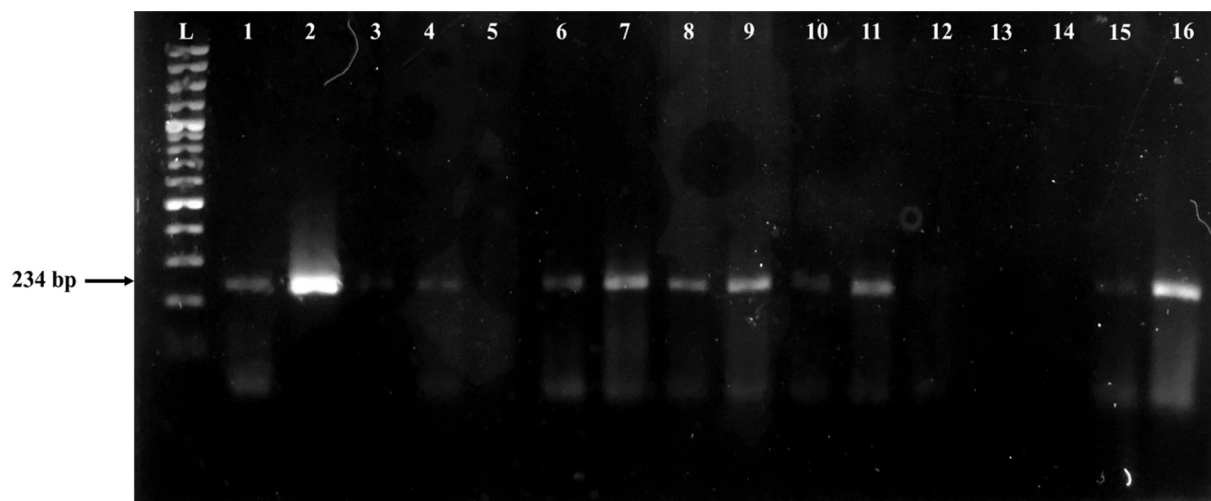


Fig. 3. Several types of weeds acted as alternative hosts of BCMV around the groundnut field. (a) *Ageratum conyzoides*; (b) *Arachis pintoii*; (c) *Ludwigia octovalvis*; (d) *Melochia corchorifolia*; (e) *Synedrella nodiflora*; (f) *Erechites valerianifolia*; (g) *Acalypha indica*; (h) *Ocimum gratissimum*; (i) *Asystasia gangetica* ssp. *micrantha*

Plant viruses can survive on alternative hosts, including weeds in cultivated and non-cultivated areas when the primary host is unavailable. Weeds can be alternative hosts for plant viruses, pathogen reservoirs, and vectors according to Wisler & Norris (2005). Our study has demonstrated that BCMV can naturally infect weeds and indicates the presence of alternative hosts in the field. BCMV has been reported to be able to infect weeds such as *Euphorbia hirta* L., *Centrosema pubescens* Benth., *Cassia tora* L., *Crotalaria mucronata* Desv., and *Crotalaria incana* L. by mechanical inoculation (Venieri 2016; Budiono, 2001), indicating their potential as alternative hosts in the field. Some other weed species that have been reported as alternate hosts of other viruses include dodder (*Cuscuta* L. spp.), which has been reported as alternate hosts of CMV, and *Sida* L. spp., *Datura stramonium* L., *Malva Tourn. ex L. spp.*, *Solanum nigrum* L., *Cleome viscosa* L., *Croton lobatus* L., and *Physalis*

L. spp., which have been reported as alternate hosts of *Tomato yellow leaf curl virus* (TYLCV) in the Dominican Republic (Salati, et al., 2002).

The RT-PCR detection confirmed the presence of BCMV-PSt in the weeds (Fig. 4), indicating their crucial role as reservoir hosts in the field. Furthermore, BCMV-PSt and *Cowpea polerovirus 2* (CPPV2) were shown to have positively infected groundnut and aphids (Fig. 5 and Fig. 6). However, BCMV-BIC, CMV, and *chickpea chlorotic stunt Polerovirus* (CpCSV) were not detected in the sample plants (Table 7). Groundnut virus is mainly transmitted by vector insects such as aphids, and seeds. As aphids play an important role in spreading the virus in the field, their control is essential. BCMV strain *Pst* is a seed-borne and non-persistent aphid-transmitted virus (Morales & Castaño, 1987), which is quite difficult to control (Sutic et al., 1999; Udayashankar et al., 2010).



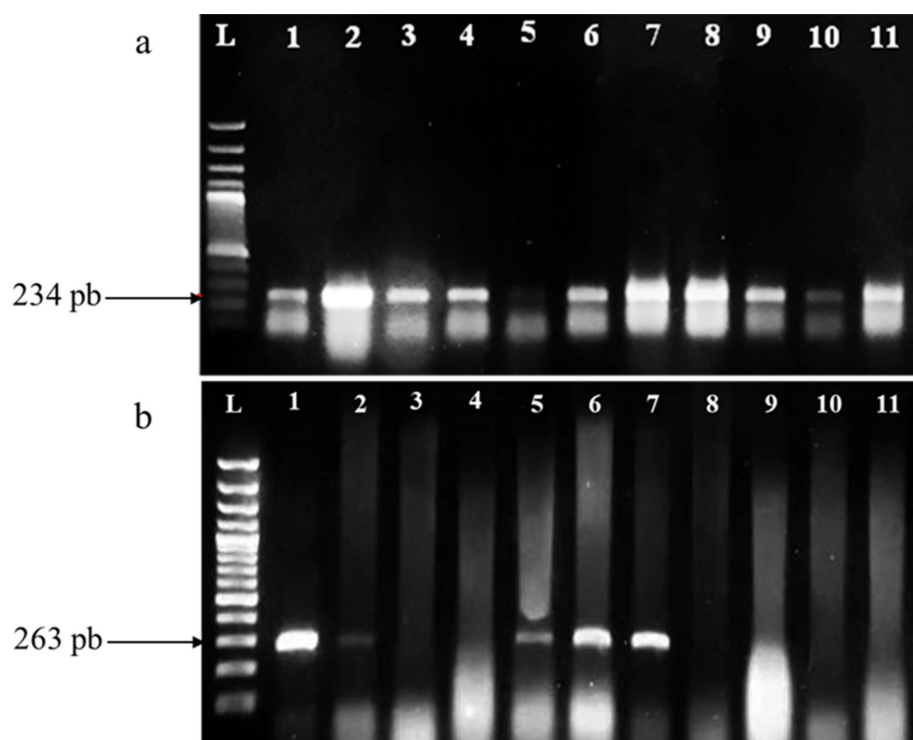
Remarks: (L) DNA ladder 100 bp (SMOBIO Technology, Inc. Taiwan); Positive control (1); Weed samples (2–16)

Fig. 4. Visualization of BCMV strain PStV PCR results for weeds in the vicinity of the groundnut field trial

Table 7. RT-PCR detection of viruses infecting groundnut, weeds, and aphids in the experimental field

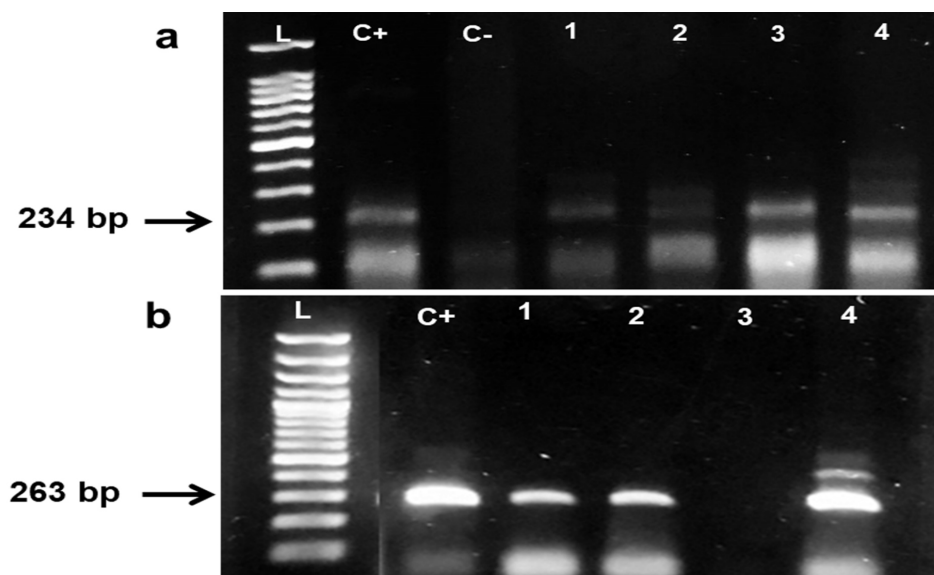
Sample	Virus frequency detected*				
	BCMV-BIC	BCMV-PStV	CMV	CpCSV	CPPV-2
Groundnut	0/10	10/10	0/10	0/10	4/10
Weed	0/16	14/16	0/16	0/16	0/16
Aphid	0/4	4/4	0/4	0/4	3/4

Remarks: * number of positive samples by RT-PCR/number of composite samples tested



Remarks: (L) DNA ladder 100 bp (SMOBIO Technology, Inc.Taiwan); positive control (1), sample M0P1 (2), M0P2 (3), M0P3 (4), M0P4 (5), M0P5 (6), M1P1 (7), M1P2 (8), M1P3 (9), M1P4 (10), M1P5 (11)

Fig. 5. PCR product of (a) BCMV strain PSt and (b) CPPV2 detected from sample plants in the field trial



Remarks: (L) DNA ladder 100 bp (SMOBIO Technology, Inc.Taiwan); positive control (C+); negative control (C-); DNA viruses from aphids (1–4)

Fig. 6. PCR products of (a) BCMV strain PStV and (b) CPPV2 on *Aphis craccivora* collected in the groundnut field trial

CPPV2, a member of the *Polerovirus*, has been reported to be transmitted by aphids. Detecting CPPV2 and BCMV in *Aphis craccivora* in this study suggests that they act as vectors for both viruses. Over 50% of seed-borne viruses are transmitted by aphids. Aphid transmission can be rapid, occurring within seconds as the aphid feeds on cell sap. The use of insecticides is often ineffective because the aphid inoculates the virus into the plant before the lethal dose of insecticide kills it. This explains why disease incidence remained high in the M0 plot despite the use of insecticides in all sub-treatments. In some cases, insecticides can even increase virus spread through an increase in vector activity (Ghewande et al., 1993; Ghewande & Nandagopal, 1997; Hooks & Fereres, 2006).

To reduce the severity of plant diseases, it is necessary to optimize cultivation factors and control aphid vectors. Epidemiological studies have provided an overview of crop management factors that should be considered, as they may be indicators of disease severity. Cultivation factors that correlate with virus incidence include planting time and seed variety (Kone et al., 2017; Gatan, 2020; Njoki et al., 2023). Planting between June and August exposes crops to high aphid numbers, potentially increasing plant disease incidence and severity (Sastry, 2013; Sastry & Zitter, 2014). Using disease-resistant varieties is also a promising strategy for controlling virus spread (Lassiter et al., 2016).

CONCLUSION

Application of *Bougainvillea* leaf extracts in the M1 plot significantly increased the effectiveness of IMP in controlling aphid-borne viruses compared to insecticide treatments. Among the tested IMPs, M1P2 and M1P4 were the most effective in controlling aphid-borne viruses infecting groundnut in the field. These treatment combinations significantly reduced aphid population and virus disease intensity and increased groundnut yields. The IMP incorporating *Bougainvillea* leaf extract could therefore be a cost-effective and eco-friendly method of controlling aphid-borne viruses on groundnut. The effectiveness of such IMPs in controlling insect-borne viruses should be investigated in the future to expand their benefits to other legumes and vegetable crops. Reusing plastic mulch a common practice among local farmers may result in lower effectiveness in controlling aphid-borne viruses than using new

plastic mulch. However, further research may be needed to prove this.

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